

Complements of Operational Research

8.2 Multi-criteria decision making: quantitative attributes and aggregation

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When there are numbers: direct and indirect attributes

For cost, range and payload there is no need to ask for opinions: the measured values give the local priorities.

Direct attribute (more is better)

$$p_i = \frac{v_i}{\sum_k v_k}$$

range, payload, yield, ...

Indirect attribute (less is better)

$$p_i = \frac{1/v_i}{\sum_k 1/v_k}$$

costs, times, distances, ...

	A	B	C
cost (€)	38 000	44 000	33 000
range (km)	220	300	160
payload (kg)	900	1000	750

This is equivalent to a perfectly consistent pairwise comparison: for range, $a_{BA} = 300/220$; for cost, $a_{CB} = 44\,000/33\,000$.

In the tests: “the usual ranking of direct and indirect attributes”.

The local priorities of the vans

	cost (ind.)	range (dir.)	payload (dir.)	service (Saaty)
A	$\frac{1/38}{1/38+1/44+1/33} = 0.332$	$\frac{220}{680} = 0.324$	$\frac{900}{2650} = 0.340$	0.261
B	0.286	0.441	0.377	0.633
C	0.382	0.235	0.283	0.106
weight w	0.482	0.272	0.158	0.088

Each column sums to 1: on each criterion, the three vans share one “unit of merit”.

C is the cheapest, but it is the worst on range, payload and service. B is the best on the last three criteria.

Aggregation: the weighted sum

$$S_i = \sum_k w_k p_{ik}$$

contribution	cost	range	payl.	serv.	S_i
A	0.160	0.088	0.054	0.023	0.324
B	0.138	0.120	0.059	0.056	0.373
C	0.184	0.064	0.045	0.009	0.302

Decision: van **B** (0.373), ahead of A (0.324) and C (0.302).

Reading: C wins on cost, the heaviest criterion, but only by 0.046 over B; B recovers 0.056 on range and 0.047 on service.

Someone looking only at the price would choose C — the last in the AHP ranking.

The AHP is *compensatory*: a bad value on one criterion can be offset by good values on the others.

Beware of what normalisation does

B costs 33% more than C, but the cost priorities are 0.286 against 0.382: dividing by the sum **compresses** the differences.

This is not a mistake: it is the **ratio scale** assumption — “C is 1.33 times better than B on cost”. If the company feels the 11 000 € difference as much more serious, it should say so in the weight of cost — and test how far the decision holds (8.3).

Priorities by sum also depend on the set of alternatives: adding or removing an alternative changes all the columns.

Limitations to keep in mind

- Full compensation between criteria: a required minimum (e.g. range ≥ 200 km) must be imposed *beforehand*, by eliminating alternatives.
- Rank reversal: adding an alternative can swap the order of the others (Belton & Gear, 1983).
- The weights only make sense in relation to the way each criterion was measured.

With the filter “range ≥ 200 km”, C would be out straight away; the choice between A and B would stay the same.

Recipe for the test problems

1. Read the Saaty rulers and write the criteria matrix (reciprocal).
2. Weights by column normalisation; $\lambda_{\max}$, CI and CR . If $CR \geq 0.10$, the data are not consistent.
3. Local priorities: costs and times as **indirect**, yields as **direct**.
4. Score $S_i = \sum_k w_k p_{ik}$; choose the largest.
5. Check the margin between the top two: if it is small, say so (8.3).

Common mistakes

- reading the ruler the wrong way round (a mark on the right favours the criterion on the right: $a_{ij} < 1$);
- treating a cost as a direct attribute;
- normalising the rows instead of the columns;
- adding up the local priorities without weighting them.

Use 4 decimal places in the intermediate calculations: the final scores often differ only in the second decimal place.

```
import numpy as np
from ahp import prioridades      # from 8.1

def direto(v):    v = np.asarray(v, float); return v / v.sum()
def indireto(v): v = 1 / np.asarray(v, float); return v / v.sum()

C = [[1, 2, 3, 5], [1/2, 1, 2, 3], [1/3, 1/2, 1, 2], [1/5, 1/3, 1/2, 1]]
S = [[1, 1/3, 3], [3, 1, 5], [1/3, 1/5, 1]]          # service
w = prioridades(C)[0]
M = np.column_stack([indireto([38000, 44000, 33000]),
                     direto([220, 300, 160]), direto([900, 1000, 750]),
                     prioridades(S)[0]])

print(M @ w)      # [0.324 0.373 0.302] -> B
```

M is the alternatives $\times$ criteria matrix;
the score is a matrix–vector product.

Try it: impose the filter “range ≥ 200 km” by removing C and recompute.
Does the order between A and B change?

Example 3 — supplier selection (exam, 11/07/2019)

A company has to decide which supplier to select, considering the criteria: 1st raw-material cost, 2nd transport cost, 3rd procurement lead time and 4th raw-material yield per tonne.

	RM cost (ton.)	Transport cost	Procurement lead time	RM yield per ton.
Supplier 1	12 000	400	15	2
Supplier 2	6000	600	10	1.5
Supplier 3	18 000	200	20	3
Supplier 4	12 000	800	5	1

Management's preferences (Saaty's scale): C1 compared with C2: 7; C1 compared with C3: 5; C1 compared with C4: 1/3; C2 compared with C3: 1/3; C2 compared with C4: 1/9; C3 compared with C4: 1/7.

Using the usual ranking of direct and indirect attributes: are the data given by the managers consistent? Which supplier was selected? Justify.

To solve in class (II)

Same statement and same criteria as Example 3, with other data.

Example 4 — 2nd test, 01/06/2018

	RM	transp.	time	yield
F1	10 000	400	12	2.5
F2	6000	700	10	2
F3	16 000	200	20	3
F4	11 000	300	15	1

C1/C2: 1/5; C1/C3: 3; C1/C4: 1; C2/C3: 7; C2/C4: 5; C3/C4: 1/3.

Example 5 — 2nd test, 27/06/2019

	RM	transp.	time	yield
F1	11 000	1400	15	3
F2	7000	1700	13	2.5
F3	15 000	1200	23	3.5
F4	12 000	1300	18	2

C1/C2: 1/3; C1/C3: 5; C1/C4: 1; C2/C3: 9; C2/C4: 3; C3/C4: 1/5.

In each case: are the judgements consistent? Which supplier should be chosen? Compare the margins between the top two.

Exercise sheet 8 — Exercise 3 (homework)

Supplier selection with verbal judgements (MATLAB solution on Moodle)

Company XYZ assesses two suppliers, A and B, according to cost, quality and delivery time.

- Cost is twice as important as quality and three times as important as delivery time; quality is four times as important as delivery time.
- On cost, B is 3 times preferable to A; on quality, A is 5 times preferable to B; on delivery time, B is 7 times preferable to A.

Write the comparison matrices, compute the weights and the consistency ratio of the criteria, and choose the supplier.

Submission: by email to the lecturer, by the deadline given in class. All exercises: [Exercise sheet 8](#).

KEY TAKEAWAYS — 8.2

1. Measured attributes: direct $v_i / \sum v$; indirect $(1/v_i) / \sum (1/v)$. They are equivalent to consistent comparisons.
2. Final score: $S_i = \sum_k w_k p_{ik}$ — the product of the local priorities matrix by the weights.
3. The AHP is compensatory; mandatory minimums are imposed beforehand, as a filter.
4. Normalisation by sum assumes a ratio scale and compresses differences; the weight of each criterion should reflect this.

Next: 8.3 Sensitivity analysis.

References and resources

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Decision Analysis.
- Saaty, T. L., How to make a decision: the analytic hierarchy process, *European Journal of Operational Research*, 48(1), 9–26, 1990.
- Belton, V., Gear, T., On a short-coming of Saaty's method of analytic hierarchies, *Omega*, 11(3), 228–230, 1983.
- Briozo, R. A., Musetti, M. A., Método multicritério de tomada de decisão: aplicação ao caso da localização espacial de uma Unidade de Pronto Atendimento – UPA 24 h, *Gestão & Produção*, 22(4), 805–819, 2015 (Moodle Example 2).

Resources

- Exercise sheet 8. `code/python/08_ahp/08_2_attributes/atributos.py`.